PAR AND FORWARD RATES

Learning Outcome

- define par and forward rates, and calculate par rates, forward rates from spot rates, spot rates from forward rates, and the price of a bond using forward rates

Par Rates from Spot Rates

An important use of spot rates is determining par rates. A **par rate** is a yield-to-maturity that makes the present value of a bond's cash flows equal to par (100% of face value). Par rates derived for *hypothetical* government bonds with different times-to-maturity are commonly used for term structure analysis because they control for tax, trading, and other potential distortions associated with actual bonds priced at either a discount or premium. The widely cited US Treasury yield curve published each day by the US Department of the Treasury, for example, is composed of par rates. The most recently issued or on-the-run government securities introduced in an earlier lesson, however, are *actual* bonds whose yields-to-maturity are close to but not equal to par rates.

On a coupon payment date, the following equation can be used to calculate a par rate by solving for *PMT* given a sequence of spot rates $Z_1, Z_2, \ldots, Z_n$. Between coupon dates, we set the flat price, rather than the full price, equal to 100.

$$100 = \frac{PMT}{(1+z_1)^1} + \frac{PMT}{(1+z_2)^2} + \cdots + \frac{PMT+100}{(1+z_N)^N}. \qquad 2$$

This equation is very similar to Equation 1, except $PV = FV = 100$. Recall that for a bond to trade at par, its coupon rate and yield-to-maturity must be equal. So, by solving for *PMT*, we also solve for the yield-to-maturity for the bond to trade at par. This single rate, divided by 100, is the par rate *per period*.

An example illustrates the calculation of par rates given spot rates. Suppose the current one-, two-, three-, and four- year spot rates on government bonds are 5.263%, 5.616%, 6.359%, and 7.008%, respectively. These are effective annual rates.

The one-year par rate is 5.263%:

$$100 = \frac{PMT+100}{(1.05263)^1}; PMT = 5.263.$$

The two-year par rate is 5.606%.

$$100 = \frac{PMT}{(1.05263)^1} + \frac{PMT+100}{(1.05616)^2}; PMT = 5.606.$$

The three-year and four-year par rates are 6.306% and 6.899%, respectively.

$$100 = \frac{PMT}{(1.05263)^1} + \frac{PMT}{(1.05616)^2} + \frac{PMT+100}{(1.06359)^3}; PMT = 6.306.$$

$$100 = \frac{PMT}{(1.05263)^1} + \frac{PMT}{(1.05616)^2} + \frac{PMT}{(1.06359)^3} + \frac{PMT+100}{(1.07008)^4}; PMT = 6.899.$$

We can confirm that these are the par rates through the pricing formula. Using these par rates as the coupon and yield-to-maturity, the price of the two-year bond is 100:

$$5.606/(1.05606)^1 + 105.606/(1.05606)^2 = 100.$$

Likewise, the price of the four-year bond is also 100:

$$6.899/(1.06899)^1 + 6.899/(1.06899)^2 + 6.899/(1.06899)^3 + 106.899/(1.06899)^4 = 100.$$

1. The par rate for a bond may be calculated from the set of spot rates reflecting maturities up to and including the maturity of the desired par rate. Using the spot rate data shown in Exhibit 1, calculate the par rate for a three-year Canadian government bond. In this example (and subsequent examples), annual payments and compounding are assumed.

Hide Solution

Solution:

Use Equation 2 to solve for *PMT*, as follows:

$$100 = \frac{PMT}{(1+z_1)^1} + \frac{PMT}{(1+z_2)^2} + \cdots + \frac{PMT+100}{(1+z_N)^N}.$$

$$100 = \frac{PMT}{(1+0.003117)^1} + \frac{PMT}{(1+0.568)^2} + \frac{PMT+100}{(1+0.7977)^3}.$$

We can factor out *PMT* and then solve for it:

$$100 = PMT \times \left(\frac{1}{(1+0.003117)^1} + \frac{1}{(1+0.568)^2} + \frac{1}{(1+0.7977)^3}\right) + \frac{100}{(1+0.7977)}$$

PMT = 0.7952.

To convert the coupon payment to a par rate per period, we divide by 100 (i.e., 0.7952/100). Thus, the par rate equals 0.7952%.

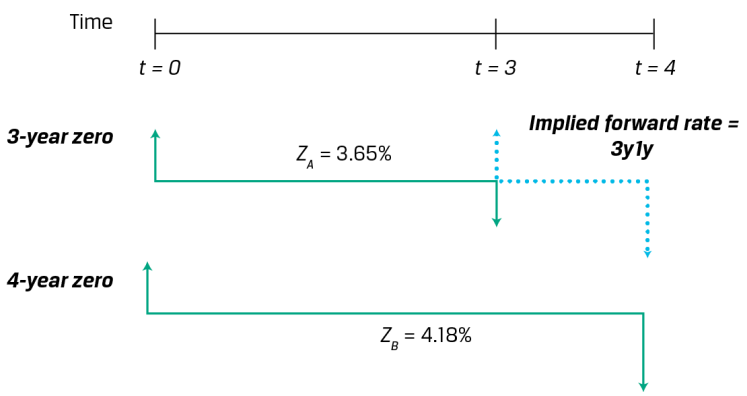
Forward Rates from Spot Rates

Spot rates can also be used to calculate **implied forward rates** (also known as forward yields), which are breakeven reinvestment rates. They link the return on an investment in a shorter-term zero-coupon bond to the return on an investment in a longer-term zero-coupon bond.

Suppose that the yields-to-maturity on three-year and four-year zero-coupon bonds are 3.65% and 4.18%, respectively. An analyst would like to know how much extra return is implied by these rates for investing for four rather than three years (i.e., an incremental year). This would be the implied one-year forward yield three years into the future, depicted in Exhibit 4.

Exhibit 4:

The One-Year Implied Rate, Three Years from Today



▶ Expand / collapse Extended Description

underlying bond, or its remaining time-to-maturity. In the money market, the forward rate is usually expressed in months.

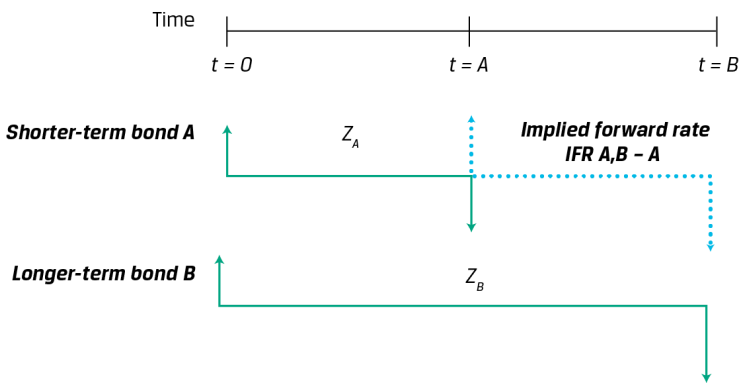
If the investor's view on future bond yields is that the one-year yield in three years' time is likely to be lower than the 3y1y, it might be better to simply invest in the four-year zero-coupon bond. However, if the investor's view is that the one-year yield will be more than the implied forward rate, the investor might prefer the three-year bond and then reinvest at the expected higher rate for an additional one year. That explains why an implied forward rate is the *breakeven reinvestment rate*.

Equation 3 is a general formula for the implied forward rate, $IFR_{A,B-A}$, for a security begins at $t = A$ and matures at $t = B$ (tenor B – A). To solve for it, we need the spot rate, z_A, and the longer-term spot rate, z_B. This is illustrated in Exhibit 5.

$$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B. \quad 3$$

Exhibit 5:

Implied Forward Rates



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Using Equation 3, let's solve for the 3y1y since we know that the yields-to-maturity on three-year and four-year zero-coupon bonds are 3.65% and 4.18%, respectively:

$$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B.$$

$$(1 + 0.0365)^3 \times (1 + IFR_{3,1})^1 = (1 + 0.0418)^4.$$

$$IFR_{3,1} = 0.057863 = 5.79\%$$

The "3y1y" implied forward yield is 5.79%. If the investor believes that the one-year yield will be higher than that in three years, she should invest in the three-year bond rather than the four-year bond (and vice versa).

EXAMPLE 3

Deriving Forward Rates from Spot Rates

1. Let's return to the spot rates shown in Exhibit 1. In the following example, we will assume annual compounding (i.e., periodicity of 1). Calculate the 2y1y implied forward rate on Canadian government bonds.

   Hide Solution

   Solution:

   To calculate the 2y1y implied forward rate, we recognize that we are estimating the one-year rate at a point two years from Time 0. Therefore, A =

$(1 + 0.00568)^2 \times (1 + IFR_{2,1})^1 = (1 + 0.007977)^3$.

$IFR_{2,1} = 0.0126$.

Thus, the implied forward rate for a one-year bond two years from now is 1.26%.

Spot Rates from Forward Rates and Bond Pricing with Forward Rates

While forward rates can be calculated from spot rates, spot rates can also be calculated from forward rates, and either can be used for pricing because they are interconnected. An example will illustrate this process.

Suppose the current sequence of one-year rates today and the one-year rate one, two, and three years from now are as follows:

Time Period	Forward Rate
0y1y	1.88%
1y1y	2.77%
2y1y	3.54%
3y1y	4.12%

These are annual rates stated for a periodicity of 1. They are effective annual rates. The first rate, the 0y1y, is the one-year spot rate. The others are one-year forward rates. Spot rates can be calculated using Equation 3. The two-year implied spot rate is 2.324%:

$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B$.

$(1 + 0.0188)^1 \times (1 + 0.0277)^1 = (1 + Z_2)^2$.

$Z_2 = 0.023240 = 2.324\%$

Notice how the product of the 0y1y and 1y1y is the two-year spot rate. This property holds in general, because the spot curve can be calculated by taking the geometric average of the forward rates. This is evident if we solve for the implied three- and four-year spot rates. Using Equation 3, we can use the implied two-year spot rate we just found to solve for the three-year spot rate:

$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B$.

$(1 + 0.023240)^2 \times (1 + 0.0354)^1 = (1 + Z_3)^3$.

$Z_3 = 0.027278 = 2.73\%$

Or since the two-year spot rate is the product of the 0y1y and 1y1y rates, we can simply multiply across the forward curve for three- and four-year spot rates:

$(1.0188 \times 1.0277 \times 1.0354) = (1 + Z_3)^3$; $Z_3 = 0.27278$

$(1.0188 \times 1.0277 \times 1.0354 \times 1.0412) = (1 + Z_4)^4$; $Z_4 = 0.030741$

Suppose that an analyst needs to value a four-year 3.75% annual coupon payment bond that has the same risks as the bonds used to obtain the forward curve. Using the implied spot rates, the value of the bond is 102.637 per 100 of par value.

$$\frac{3.75}{(1.0188)^1} + \frac{3.75}{(1.023240)^2} + \frac{3.75}{(1.027278)^3} + \frac{103.75}{(1.030741)^4} = 102.637.$$

The bond also can be valued using the forward rates as follows:

$$\frac{3.75}{(1.0188)} + \frac{3.75}{(1.0188 \times 1.0277)} + \frac{3.75}{(1.0188 \times 1.0277 \times 1.0354)} + \frac{103.75}{(1.0188 \times 1.0277 \times 1.0354 \times 1.0412)} = 102.637.$$

Forward Rates

Suppose we have the following forward rates for Canadian government bonds:

Forward tenor	Rates
0y1y	0.3117%
1y1y	0.8250%
2y1y	1.2587%
1y2y	1.0416%

1. Calculate a three-year spot rate on Canadian government bonds using the forward rate information.

   Hide Solution

   Solution:

   The three-year spot rate is calculated using three of the one-year forward rates above: the 0y1y, 1y1y, and 2y1y.

   $(1.003117 \times 1.008250 \times 1.012587) = (1 + z_3)^3$.

   $z_3 = 0.7977\%$.

2. Calculate the value of a three-year Canadian government bond paying a 0.50% coupon using forward rates.

   Hide Solution

   Solution:

   $$PV = \frac{0.5}{1.003117} + \frac{0.5}{(1.003117 \times 1.008250)} + \frac{100.50}{(1.003117 \times 1.008250 \times 1.0012587)}$$

   PV = 99.126

   This example corresponds to Question 4 in the earlier Lesson Question Set. A review of the solution to that problem indicates that we have shown that the value of the three-year 0.50% coupon Canadian government bond is equal to 99.126 regardless of whether we use spot rates or forward rates.

Question Set

1. Contrast a par rate and a yield-to-maturity.

   Hide Solution

   Solution:

   A par rate is the YTM and coupon rate assuming a bond is priced at par value and cash flows are discounted using spot rates, while the yield-to-maturity is the internal rate of return on a bond's cash flows using a bond's current price.

2. Calculate a three-year par rate given the following spot rates. Assume annual compounding:

1-year	3.0000%
2-year	2.8000%

Solution:

2.607%.

$$100 = \frac{PMT}{(1+z_1)^1} + \frac{PMT}{(1+z_2)^2} + \ldots + \frac{PMT+100}{(1+z_N)^N}$$

$$100 = \frac{PMT}{(1+0.03)^1} + \frac{PMT}{(1+0.028)^2} + \frac{PMT}{(1+0.026)^3} + \frac{PMT+100}{(1+0.026)^3}$$

$$100 = PMT \times \left(\frac{1}{(1+0.03)^1} + \frac{1}{(1+0.028)^2} + \frac{1}{(1+0.026)^3}\right) + \frac{100}{(1+0.026)^3}$$

PMT = 2.607%

3. Match the implied forward rate to the spot rates necessary to calculate it.

1. 1y2y	A. Three-year and two-year spot rates
2. 3y2y	B. Three-year and one-year spot rates
3. 2y1y	C. Five-year and three-year spot rates

Hide Solution

Solution:

1. B. The 1y2y is a two-year forward rate in one year. Thus, we start with the three-year spot rate and use the one-year spot rate to find an implied two-year forward rate.

2. C. The 3y2y is a two-year forward rate in three years. Thus, we start with the five-year spot rate and use the three-year spot rate to find an implied two-year forward rate.

3. A. The 2y1y is a one-year forward rate in two years. Thus, we start with the three-year spot rate and use the two-year spot rate to find an implied one-year forward rate.

4. Determine the missing words in the following sentence: An implied forward rate represents a _____ reinvestment rate such that no arbitrage opportunities exist between spot and forward rates.

Hide Solution

Solution:

An implied forward rate represents a breakeven reinvestment rate such that no arbitrage opportunities exist between spot and forward rates.

5. Calculate the implied forward rate 2y3y given the following spot rates. Assume annual compounding.

Tenor	Spot Rate
1-year	2.0%
2-year	2.3%
3-year	2.5%
4-year	2.6%
5-year	2.4%

Hide Solution

Solution:

2.467%.

$Z_3 = 2.487\%$

6. Calculate the price of a 2% coupon bond with 100 par value maturing in three years given the following forward rates, and demonstrate that this price is equivalent regardless of whether spot rates or forward rates are used as discount rates. Assume annual compounding and the following forward rates.

Tenor	Forward Rate
0y1y	1.5%
1y1y	2.5%
2y1y	3.5%

Hide Solution

Solution:

The price is 98.619.

Forward rate approach:

$$PV = \frac{2}{(1.015)} + \frac{2}{(1.015 \times 1.025)} + \frac{102}{(1.015 \times 1.025 \times 1.035)}.$$

$PV = 98.6190$.

The implied one-, two-, and three-year spot rates are as follows:

One-year = 1.5%.

Two-year = $(1.015 \times 1.025)^{(1/2)} - 1 = 1.999\%$.

Three-year = $(1.015 \times 1.025 \times 1.035)^{(1/3)} - 1 = 2.4967\%$.

Using the spot prices to price the bond gives the following:

$$PV = \frac{2}{(1.015)^1} + \frac{2}{(1.01999)^2} + \frac{102}{(1.024967)^3}.$$

$PV = 98.6190$.

7. Demonstrate how an investor with a five-year investment horizon would optimally decide between the two following strategies:

- Buy a three-year government bond at the 3% spot rate, and then reinvest for two years at the expected future two-year rate.
- Buy a five-year government bond at the 4% spot rate.

Hide Solution

Solution:

Calculate the implied two-year forward rate in three years:

$$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B.$$

$$(1 + 0.03)^3 \times (1 + IFR_{3,2})^2 = (1 + 0.04)^5.$$

$$IFR_{3,2} = 5.52\%$$

If the investor believes that the two-year rate in three years will be greater than 5.52%, then the better strategy is to buy the three-year government bond and reinvest at a higher rate. If the investor believes that the two-year rate in three years will be less than 5.52%, then the five-year bond should be purchased. The implied forward rate represents the breakeven reinvestment rate.

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DR

looking at the formulas makes me want to rip my hair out

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AS

this section is a nightmare...so many mistakes

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MK

There is a mistake in calculating the three year spot rate where it says,

(1+0.023240)^2 * (1+ 0.0277) = (1+Z3)^3.

The implied forward rate for this should be 2y1y not 1y1y. Thus, the correct way of calculating this would be:

(1+0.023240)^2 * (1+ 0.0354) = (1+Z3)^3.

Created 4 months ago by MINJAE KIM — 2 replies | Last Activity: 3 months ago

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SG

On this example

(1 + ZA)^A × (1 + IFRA,B – A)^(B – A)= (1 + ZB)^B

(1 + 0.0188)^2 × (1 + 0.0277)^1 = (1 + Z2)^3

Z2 = 0.023240 = 2.324%

I think the result is incorrect because 1.0188^2 x 1.0277^1 = 1.066705 and the cube root will be 1.021758 = 2.1758% which I believe is not the spot rate

I obtain 2.324% with 1.0188 x 1.0277 and then the square root of the result. If you capitalize 100 two years at a 2.324% is the same result as capitalizing on the first year at 1.88% and reinvest on a second year at a yield of 2.77%

Created 5 months ago by Sascha Gunter Mas Weber — 4 replies | Last Activity: 3 months ago

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MO

Where did they get the rates for Exmaple 2? They're saying Exhibit 1? But Exhibit 1 has no mention of any rates? Do they mean Exhibit 3? But the rates in Exhibit 3 are not what they're using in their solution?

Honestly the amount of mistakes in these modules is staggering. I am getting really tired of figuring stuff out...

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LA

Agreed

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DA

Typo in EXAMPLE 4 Q2:

Should be 1.012587 not 1.0012587 for 2y1y

Created 4 months ago by Dhevin Albel Singh Gill — 1 reply | Last Activity: 4 months ago

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MS

From my understanding, par rates are derived based on spot rates. In a previous lesson, it was mentioned that the ideal spot curve consists of the yields to maturity (YTMs) of default-risk-free zero-coupon bonds. However, in practice, zero-coupon bonds do not exist for all maturities needed to construct a complete spot curve. Therefore, I would like to clarify: how are spot rates actually derived in practice when zero-coupon bonds are not available, and how are these estimated spot rates then used to calculate the par rates?

Created 8 months ago by Marko Simovic — 1 reply | Last Activity: 5 months ago

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SH In Example 3, if B= 3 and A=1, then why is B-A = 1, should A not be equal to 2 since we are calculating 2y1y?

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